

IOT BASED HEART MONITORING AND RISK PREDICTION

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Title Justification

- The title clearly reflects the project's focus on continuous heart monitoring using IoT sensors.
- It highlights the risk prediction capability, which classifies the user's heart condition into meaningful levels.
- The title is simple, direct, and descriptive, making the project's purpose easy to understand.
- It emphasizes both real-time data collection and intelligent decision-making, aligning with the system's objectives.



Abstract

Heart diseases are increasing globally due to lifestyle changes, stress, and the lack of continuous cardiac monitoring, while traditional systems fail to provide real-time tracking and accessible long-term observation.

To address this gap, the proposed work develops an IoT-based heart health monitoring and risk estimation system capable of continuously measuring key cardiac parameters such as ECG signals, heart rate, and SpO₂.

The system processes the acquired data using a lightweight rule-based algorithm to classify the user's condition into low, medium, or high risk levels, and automatically stores the values on a cloud platform for easy access.

A dashboard is included for simple visualization of cardiac trends. Experimental observations show that the system reliably captures physiological signals and provides meaningful risk indications, making it a low-cost, effective solution for remote monitoring and early detection of heart abnormalities.



Problem Statement

- Traditional heart monitoring systems are expensive, non-portable, and do not support continuous long-term tracking, leading to delayed detection of cardiac abnormalities.
- Most existing IoT-based systems only display raw physiological data without meaningful analysis or risk prediction, making it difficult for users to understand their heart condition.
- There is a need for an affordable IoT platform that can continuously collect cardiac data, analyze it using AI-assisted logic, and store it in the cloud, enabling early detection and remote monitoring.

Objectives of Project

Objective-1: IoT-Based Monitoring

Develop a system that continuously monitors essential cardiac parameters to support early detection of abnormalities in heart health.

Objective-2 : AI-Assisted Analysis

Design and implement an AI-assisted rule-based model that analyzes real-time cardiac data and classifies heart condition into meaningful risk levels through cloud-based visualization.

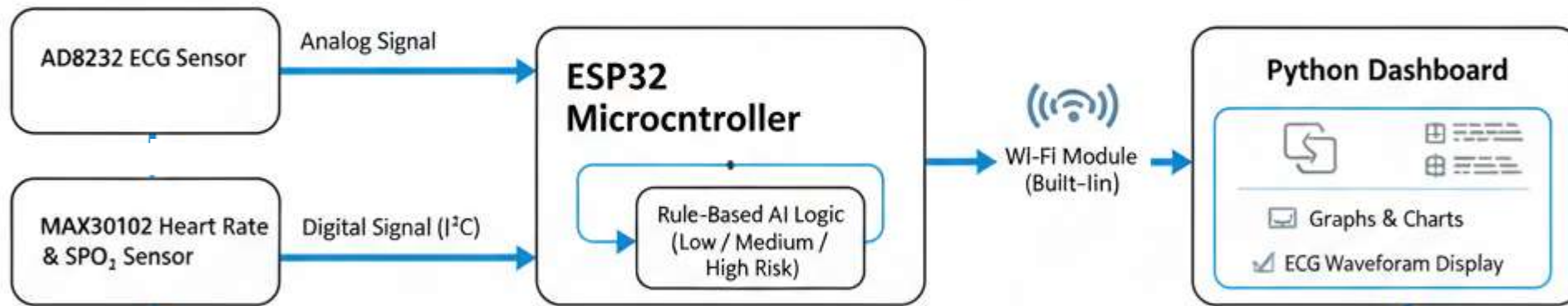
Literature survey for objectives

Author(s) / Source	Year	Objective Linked	Main Contribution	Research Gap Identified
PhilStat – <i>IoT-Based ECG Monitoring System for Health Care Applications</i>	2022	Objective 1	Authors developed an IoT-based ECG monitoring system capable of transmitting ECG signals for basic remote health monitoring.	Only raw ECG data is transmitted; no cloud storage, no dashboard, no trend visualization.
IJRASET – <i>IoT-Based Smart ECG Monitoring System</i>	2023	Objective 1	Authors implemented a low-cost ECG monitoring prototype providing real-time ECG display on IoT hardware.	Provides only basic real-time display; lacks long-term storage, analytics, and user-friendly visualization.
Kim et al. – <i>TinyML-Based Classification in an ECG Monitoring Embedded System</i>	2023	Objective 2	Authors demonstrated that TinyML can run ECG classifiers directly on embedded devices with reduced memory and power usage.	Does not provide simple interpretable rule-based risk levels; still computationally heavier than basic embedded logic.
N. Divya, M. Riyazuddin, A. Ahad, S.& R. Vulapula,, “ <i>Predicting Heart Disease Using Machine Learning and IoT Techniques,</i> ”	2024	Objective 2	The authors propose a ml and IoT-based framework that collects patient physiological and contextual data, then uses multiple classifiers (SVM, Naïve Bayes,) to predict heart disease risk.	Does not present an embedded, low-compute rule-based model for real-time microcontroller operation, nor does it clearly provide cloud dashboard visualization with risk levels.



Proposed System

IOT-BASED HEART DISEASE MONITORING AND RISK PREDICTION SYSTEM



Sensors → ESP | Processing → Cloud Storage → Dashboard Vissalization

Proposed System

Module 01

Data Acquisition

AD8232 ECG sensor captures electrical cardiac activity.
MAX30102 sensor measures heart rate and SpO₂.

Module 02

Processing & Classification

ESP32 microcontroller performs on-device processing.
Lightweight AI rule-based algorithm classifies condition into Low, Medium, or High risk.

Module 03

Cloud Storage

Data transmitted via Wi-Fi to Google Sheets for real-time storage and remote accessibility.

Module 04

Visualization

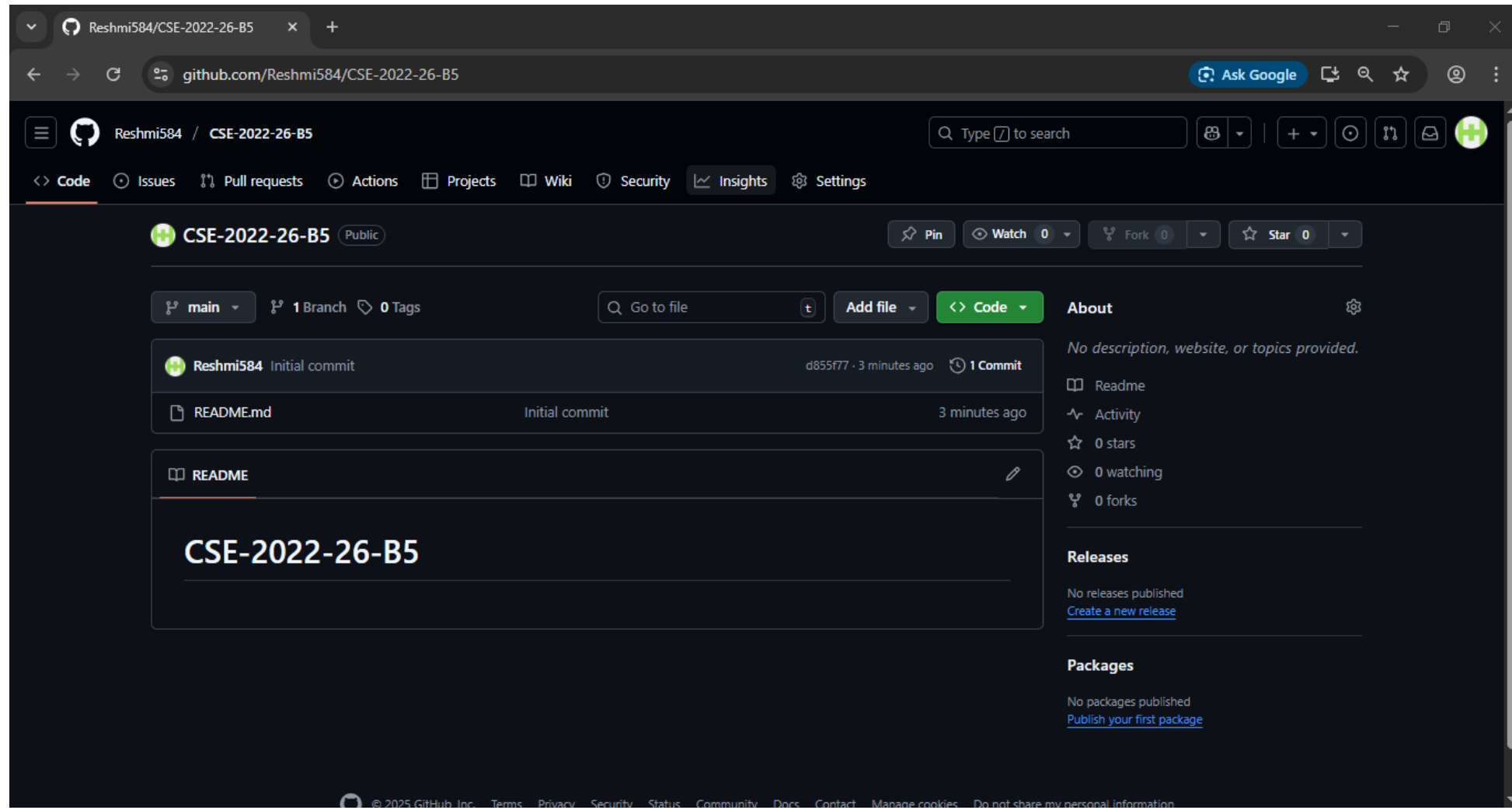
Dashboard displays visual trends, ECG patterns, risk indicators, and interpreted summaries.

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Git Hub Dashboards of each student



<https://github.com/Reshmi584/CSE-2022-26-B5>

Any Queries?

